# Project INSPIRE Design Document

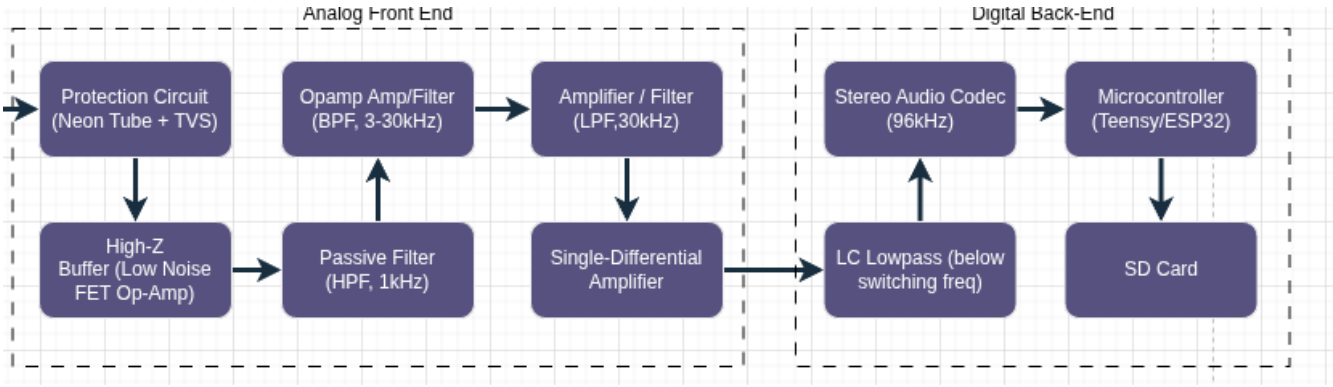
## Project Part III: High-Pass Filter

Revision 1.0: Presented 10/9/2026

## 1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:




The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

## 2) Sub-circuit Specifications

The Passive High-Pass Filter follows the High-Z buffer to block out intense, low-frequency hum from power grid  harmonics. Because these power line signals are massive emitters of interference, filtering them out immediately after buffering prevents downstream amplifiers from clipping and preserves the dynamic range of the receiver.

This block implements a second-order passive RC high-pass filter to achieve a sharper, steeper roll-off (-40 dB/decade) than a single-stage filter, establishing a sharp cutoff frequency at 1kHz.

- Filter Topology: The sub-circuit will use a cascaded, two-stage resistor-capacitor (RC) network. Because it is completely passive, it requires no power rails, introducing zero active electronic noise into the ultra-sensitive VLF front-end.
- Impedance Matching & Loading: To prevent the second RC stage from loading down the first RC stage, which distorts the filter's transfer function and dampens the corner frequency), the component values must be scaled. The second stage's resistor value should ideally be at least 10 times larger than the first stage's resistor.
- Component Selection: Capacitors must be selected for low leakage, high stability, and tight tolerances (e.g., film or C0G/NP0 ceramic capacitors) to prevent drift in the 1kHz cutoff threshold. Resistor values must be balanced to minimize thermal Johnson noise while remaining low enough not to be loaded down by the next active stage.

## 3) Expected Outputs

This sub-circuit design requires a technical write-up and simulation covering two major parts:

- **Mathematical Derivation & Component Values**: Theoretical calculations for the component values required to achieve a 1kHz corner frequency with a second-order passive response, including a short paragraph justifying the choice of impedance scaling factor.

- **SPICE Simulation**: A SPICE simulation of the passive filter network. The simulation must display:
    - An AC Bode Plot (Magnitude and Phase): Spanning from below the cutoff to above the cutoff by at least an order of magnitude both ways. This way you can explicitly show the roll-off and behavior.
    - Loading Analysis: A short paragraph showing how the filter's frequency response changes when subjected to another load resistor on the output. If this is detrimental to your desired response, mention if a op-amp buffer is required.

***Optional Extra (Stretch Goals):***

- Add component tolerances and parasitics into the schematic to simulate 1% or 5% component variation effects, and how real components will change the response.